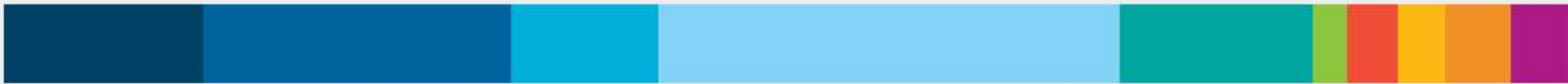




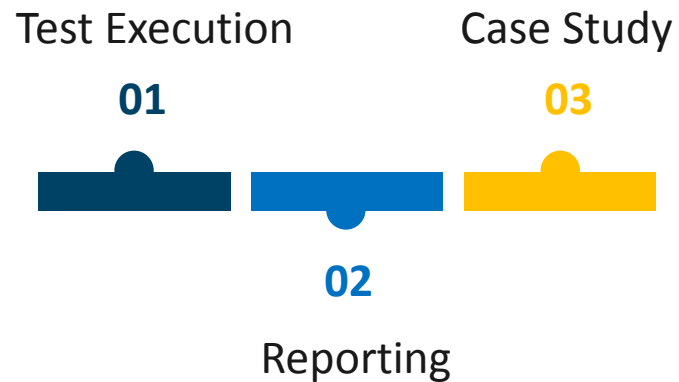
Basic Testing



Days 7 and 8: Test Execution and Reporting
(with Case Study)

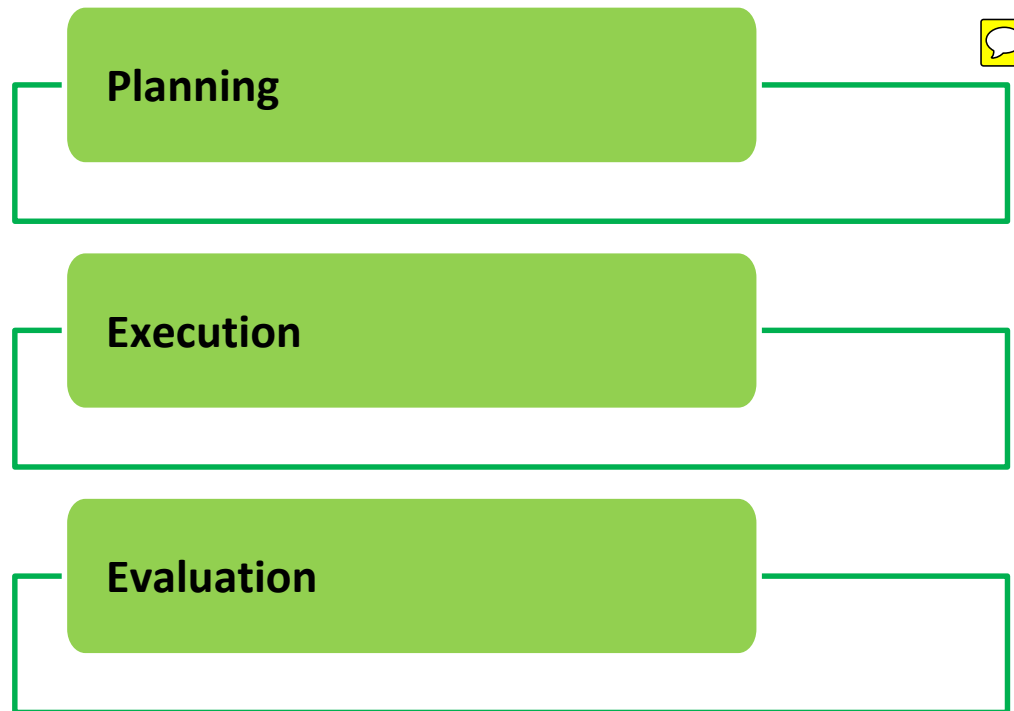


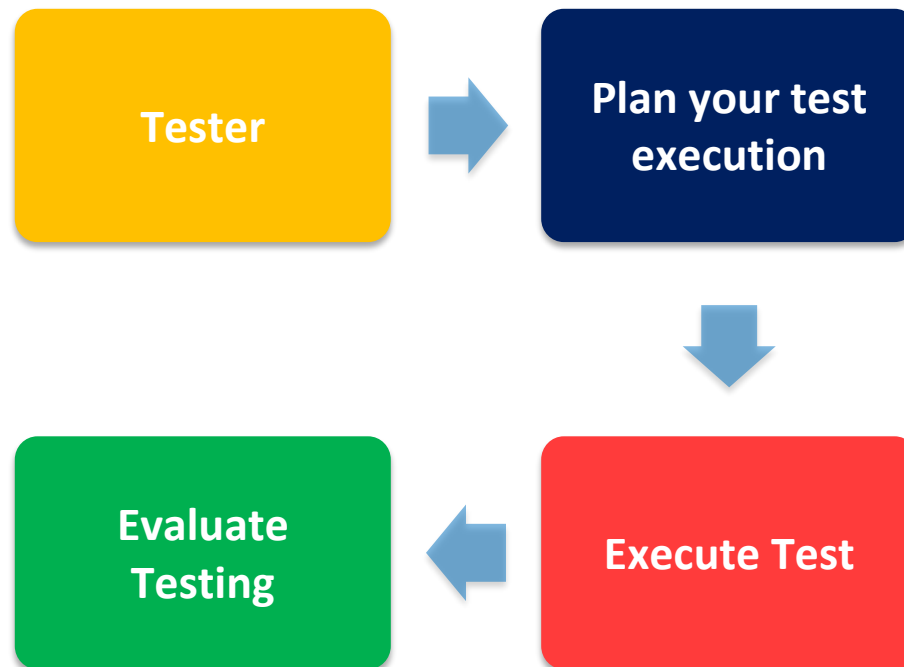
Day 07 and 08



01 Test Execution

A single test normally includes three steps:





General test procedure (continued)



Plan Your Text Execution

If I click OK, without input into text field, the application will probably crash.

Execute Test

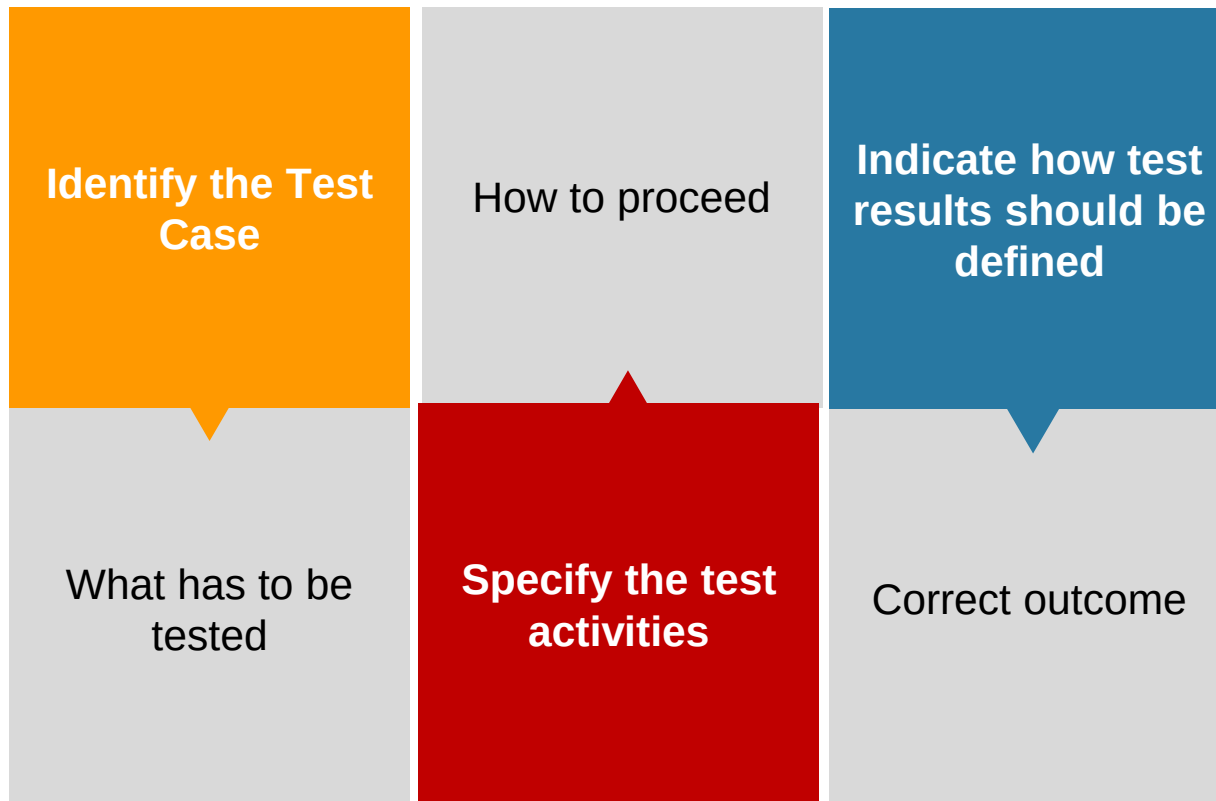
Let's see if it does.

Evaluate Testing

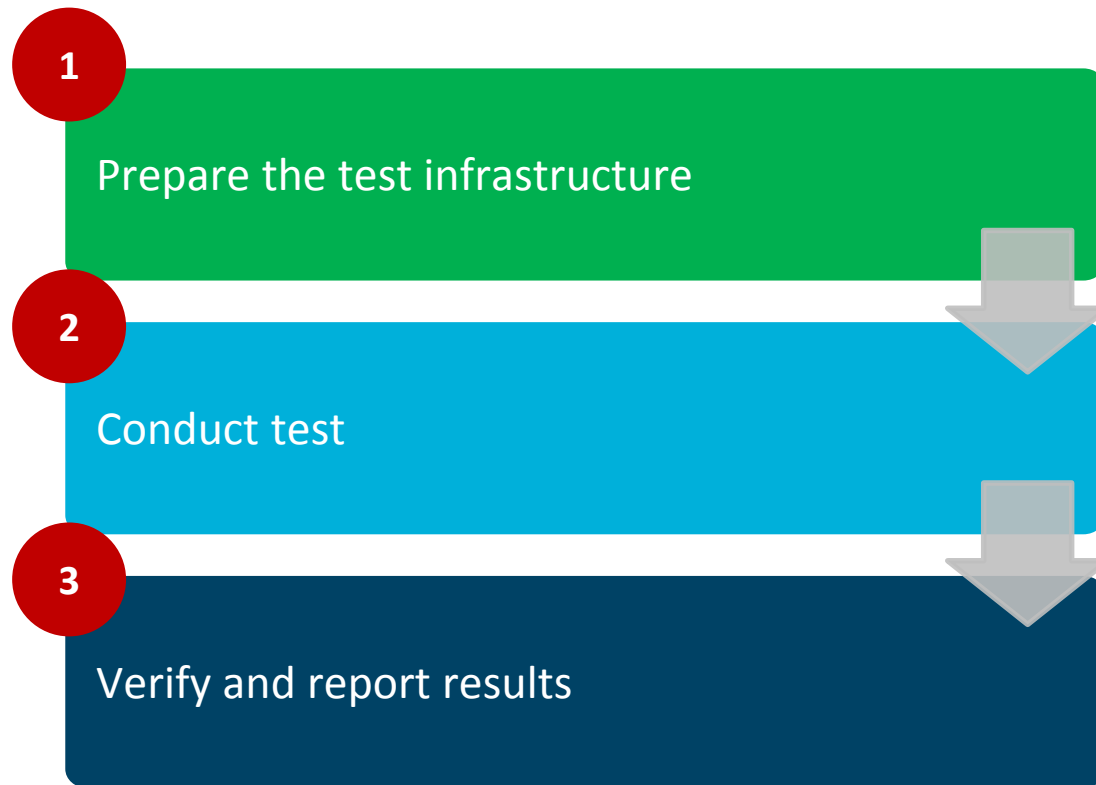
It sure did!

Planning

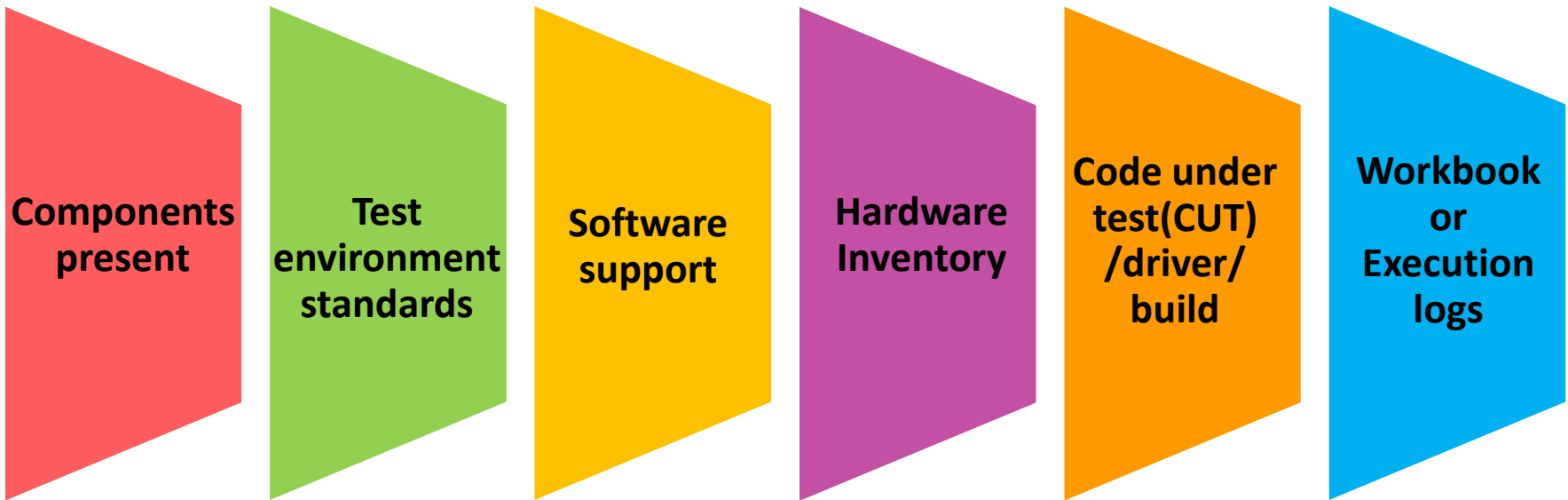
- It is the first step to determine what should be tested-- also referred to as test case design
- Design the test cases (inputs and outputs) used to test the system



Execution of a test is a **three**-step process:



Prepare test infrastructure



Test Effectiveness

- Adequate test environments are vital towards test effectiveness, especially in the area of non-functional testing.

Timely Availability

- It is important that a test environment is timely available for efficient testing.

Appropriate Environment

- Development environment
- Testing environment
- Acceptance environment
- Production environment

Simulate Adequately

- Testing conditions simulate production conditions.

Tailored to Need

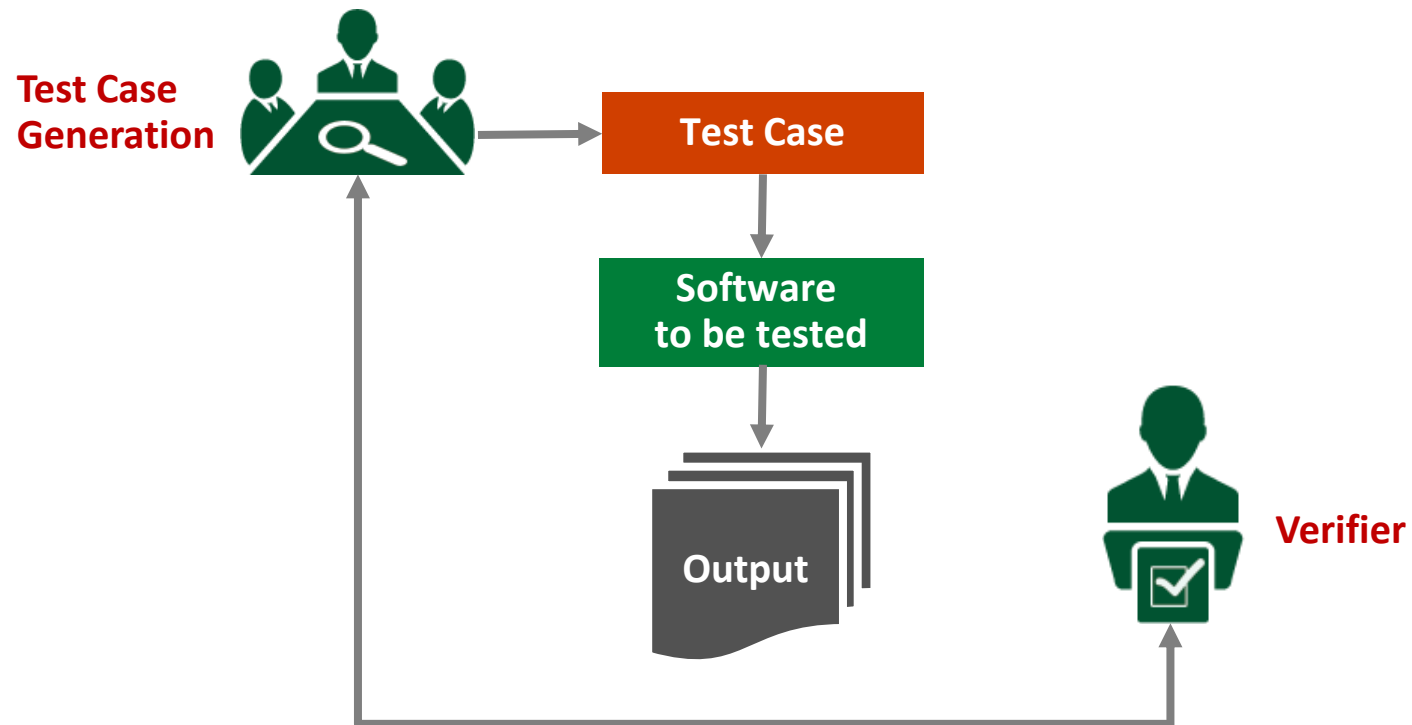
- Test environments cater to specific test needs.

Execute test case

- The purpose of the task is to implement the test execution plan, produce test results, and compare between the expected test results and actual test results.
- Execute each test exactly as defined in the test-case implementation.

Execute test case

The process to execute a test case is shown below.



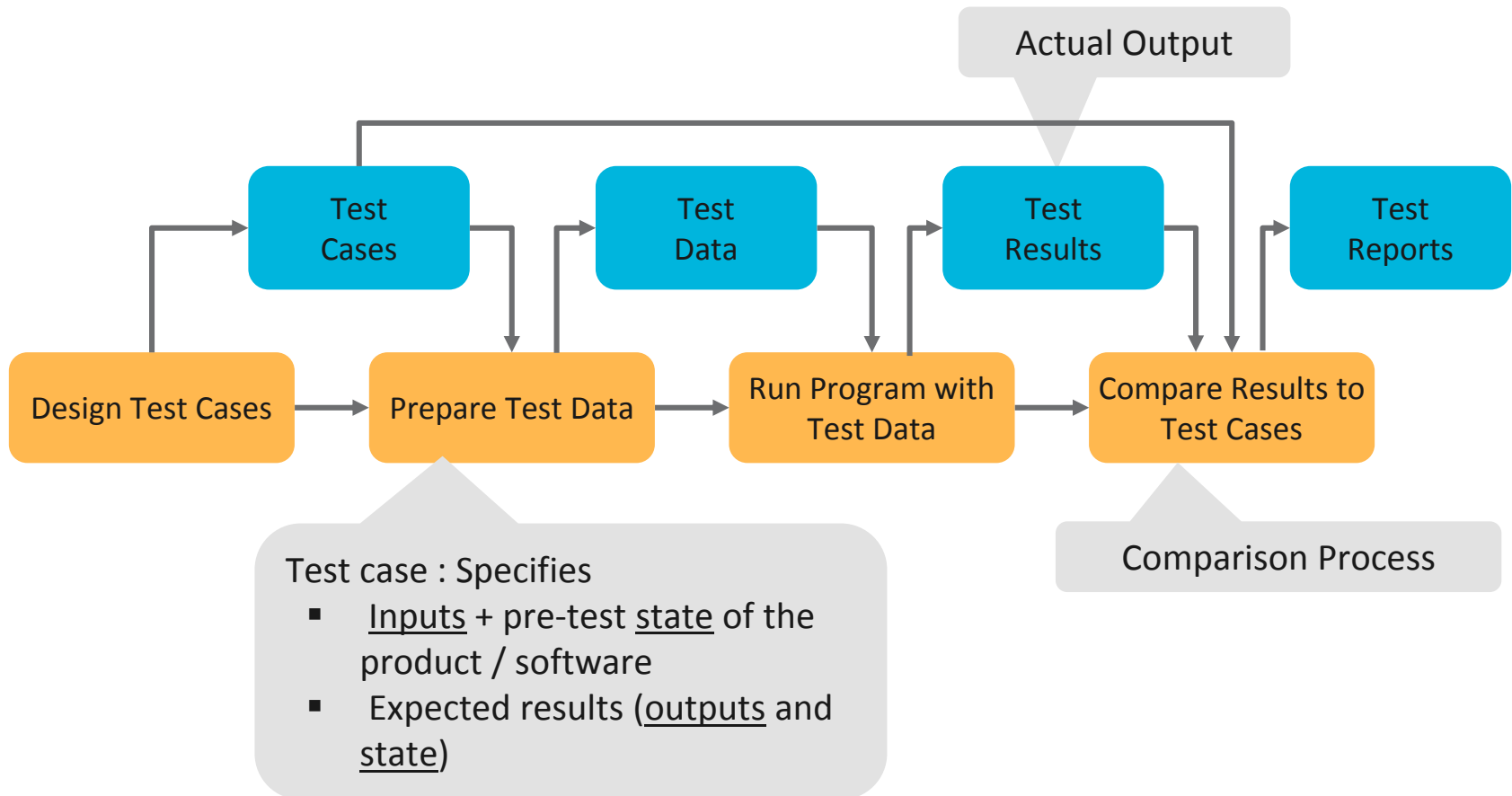


Questions?

02 Reporting

Test Execution Status Report

- This is a communication sent out to establish transparency to the QA team's activities of the day during the test cycle.
- This includes both defect information and test case run information.
- It is sent to Development, Environment support, Business analyst, and the Project teams.



Only exhaustive testing can show a program is free from defects.
However, exhaustive testing is impossible.

Testing policies define the approach to be used in selecting
system tests:



All functions should be tested.



Where user input is required, all
functions must be tested with correct
and incorrect input.

Testing guidelines are hints for the testing team to help them choose tests that will reveal defects in a system.

Enter all positive / correct data to validate the happy path of testing.

Choose inputs that force the system to generate all error messages:

- Design inputs that cause buffers to overflow
- Repeat the same input or input series several times
- Force invalid outputs to be generated
- Force computation results can be too large or too small

Verify the
existence of
the defect

- Verify that the element under test, the execution environment, and the test case-implementation are consistent with the test case design.
- Verify and confirm the discrepancy between the expected test results and actual test results.



The purpose of the task is to analyze and resolve variances encountered during test execution.

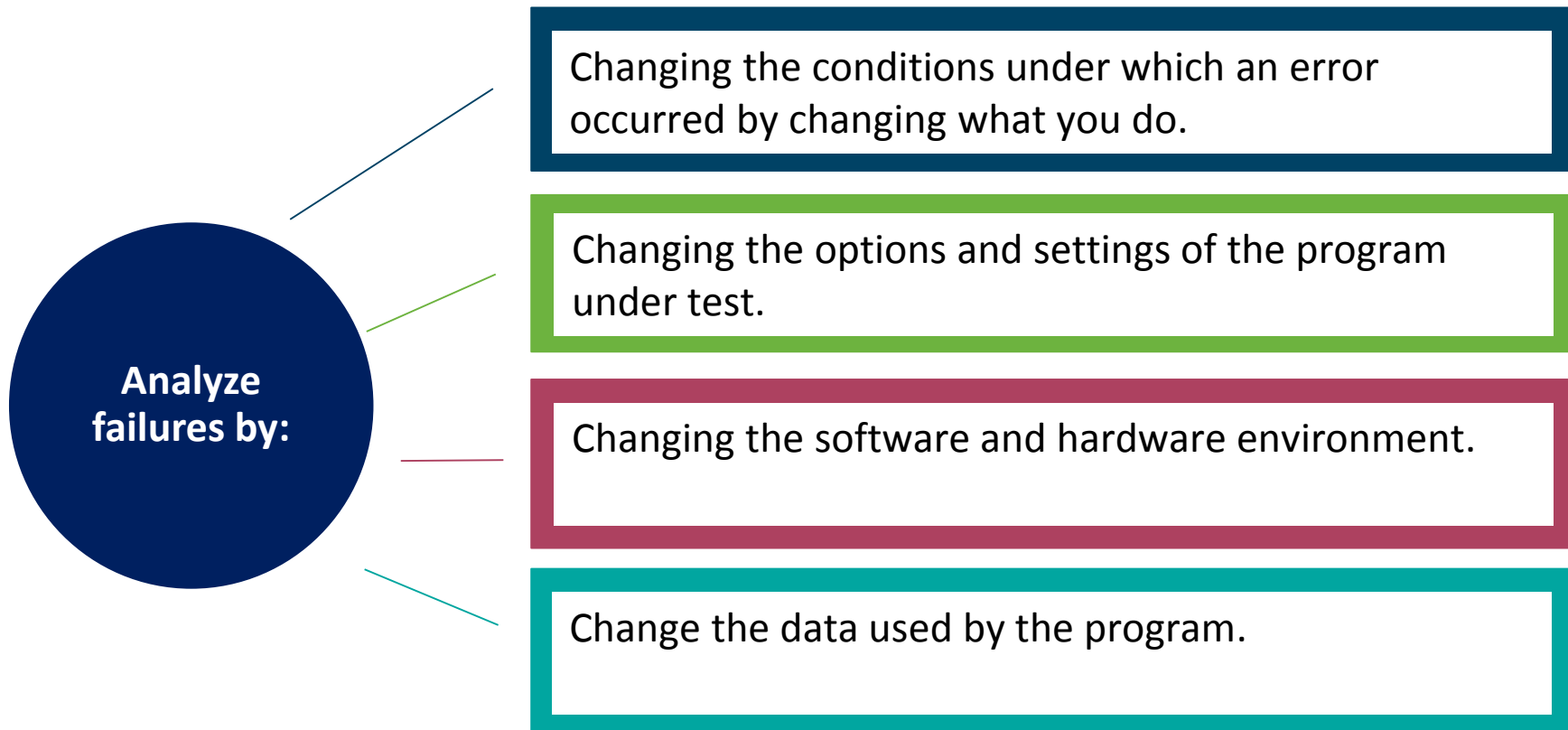
Each variance is analyzed to determine the cause.

Analysis may indicate that the cause of the variance is a defect, or that a change request is required.

Investigate further and gather as much information as possible about the symptoms.

Analyze failures by varying conditions, options, and settings to gauge the severity of the defect.

Analyze Failures by varying Conditions, Options, and Settings





Questions?

03 Case Study

Case Study: Day 8—Test Case Execution and Reporting



Let us get started with some real life case studies now. Here is what you need to do:

- Work with your team as per instructions from the facilitator
- Assign the test cases to the tester as per the test plan in RQM
- The testers would execute the application using the test cases and record the defects in RQM
- The observer would observe the defect recording style of testers
- Share your key takeaways with the class on the steps you took to prepare the test cases (30 mins)



Microsoft Word
17 - 2003 Document



Questions?